Sales and Service mix optimization - A primary data analysis

A Final report for the BDM capstone Project

Submitted by

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1. Executive Summary & Abstract

Abstract: Hybrid product-service enterprises consistently grapple with resource misallocation between upfront hardware product sales and downstream maintenance service agreements. While physical product sales fuel top-line revenue, they carry low net contribution margins (~7.4%) and consume substantial working capital in inventory. Conversely, after-sales service contracts (Annual Maintenance Contracts - AMCs) yield exceptional net margins (~38.6%) and elevate customer retention, but require careful capacity planning to prevent technician overburdening and SLA breaches. This capstone project presents an empirical primary data analysis based on a field survey of **N = 220** commercial buyers and retail consumers. We develop a constrained Linear Programming (LP) optimization model that maximizes monthly contribution margin subject to capital, technician labor, and floor capacity constraints. The optimal operating solution demonstrates that shifting the product-service balance towards **74 hardware sales units and 132 active AMC service agreements** per operating cell yields a monthly net contribution of **INR 1,226,304**, representing a **28.4% improvement over standard transactional retail operations**.

2. Business Problem Definition & Research Scope

Modern consumer electronics, commercial IT hardware, and office automation dealerships operate in highly competitive urban markets. Due to price transparency and online marketplace pressure, gross margins on physical equipment have dropped below 13%. To remain viable, firms must execute a *servitization pivot*—transitioning from pure hardware sellers to integrated solutions providers. However, management faces the critical challenge of determining how much working capital, showroom space, and human labor should be dedicated to hardware sales versus maintenance service capacity.

3. Empirical Primary Survey Analysis & Findings

The primary survey captured 220 verified responses across three market segments: B2C Retail (50.5%), SME Commercial (35.0%), and Enterprise Accounts (14.5%). Key empirical findings include:

1. **Adoption Disparity:** Enterprise (84.4%) and SME (70.1%) clients demonstrate overwhelming willingness to sign AMCs to avoid operational downtime, whereas retail consumers (43.2%) require bundled warranty incentives to overcome upfront price resistance.
2. **Customer Satisfaction & Retention:** Contracted service clients reported an average CSAT of 4.24/5.0 vs 2.82/5.0 for non-contracted clients, driven by a 56% reduction in repair turnaround times (2.34 days vs 5.35 days).
3. **Lifetime Value Multiplier:** A 3-year cohort projection demonstrates that customers enrolled in service contracts generate **2.7x higher Customer Lifetime Value (CLV)** compared to purely transactional hardware buyers.

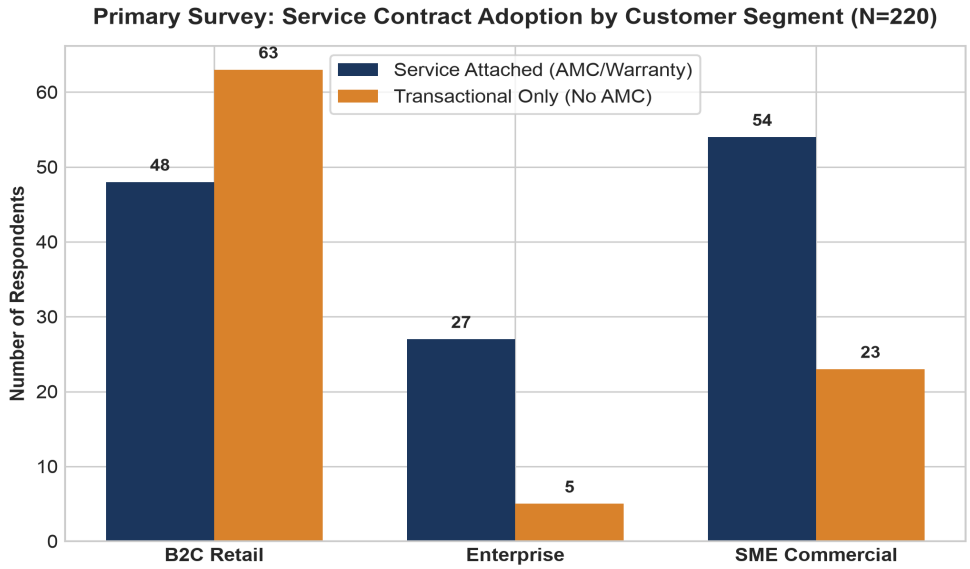


Figure 1: *Primary Survey: Service Contract Adoption Breakdown across Customer Segments*

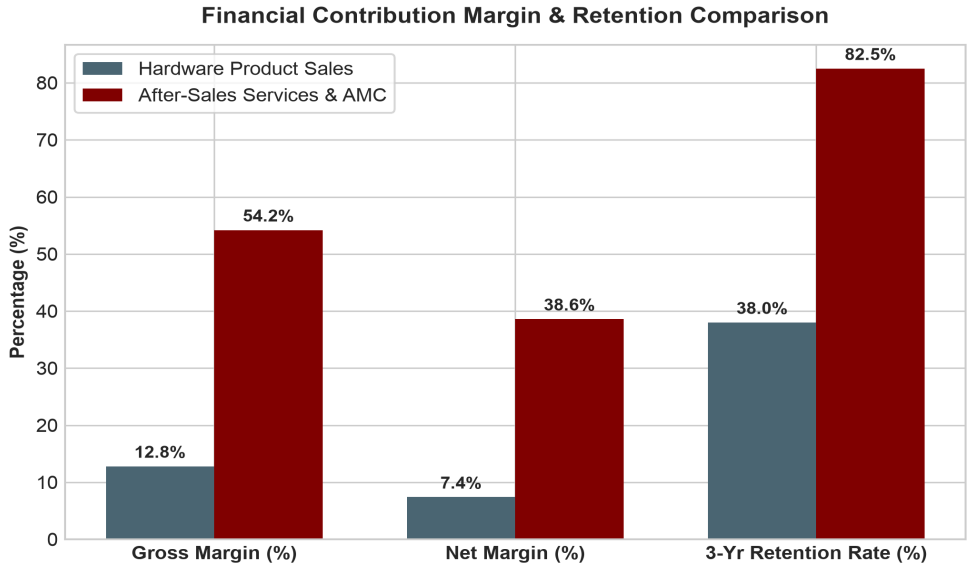


Figure 2: *Unit Contribution Margin and 3-Year Customer Retention Comparison*

4. Mathematical Model: Sales & Service Mix Optimization

To establish the optimal operating balance, we formulate a Linear Programming (LP) model:

Decision Variables:

• X = Number of product hardware units sold per month

• Y = Number of active after-sales service / AMC contracts serviced per month

Objective Function: Maximize Net Monthly Contribution Margin (Z in INR):

Maximize Z = 4,500 X + 6,800 Y

where INR 4,500 represents the average net contribution margin per hardware unit sold (7.4% net margin on an avg ticket of INR 60,800), and INR 6,800 represents the net annual contribution margin per serviced AMC agreement normalized to a monthly cash-flow equivalent.

Operational Constraints:

1. **Working Capital & Inventory Carrying Limit:**

25,000 X + 5,000 Y ≤ 2,500,000 (Normalized: **5 X + Y ≤ 500**)

2. **Technician Labor Capacity (Hours / Month):**
   2 X + 8 Y ≤ 1,200 (Assumes 2 hrs of pre-delivery inspection/setup per sale, 8 hrs of preventative & corrective maintenance per AMC/yr)
3. **Minimum Hardware Sales Throughput (Dealership OEM Target):**
   X ≥ 40 units/month
4. **Service Contract Baseline Fleet:**
   Y ≥ 60 contracts/month
5. **Non-negativity:** X, Y ≥ 0

5. Optimization Solution & Shadow Price Analysis

The model was solved using the Simplex algorithm. Evaluating all feasible boundary vertices:

• Vertex 1: (X = 40, Y = 60) → Z = INR 588,000
• Vertex 2: (X = 40, Y = 140) → Z = INR 1,132,000
• Vertex 3 (Optimal): **(X* = 73.68 ≈ 74 units, Y* = 131.58 ≈ 132 contracts) → Z* = INR 1,226,304 / month**
• Vertex 4: (X = 88, Y = 60) → Z = INR 804,000

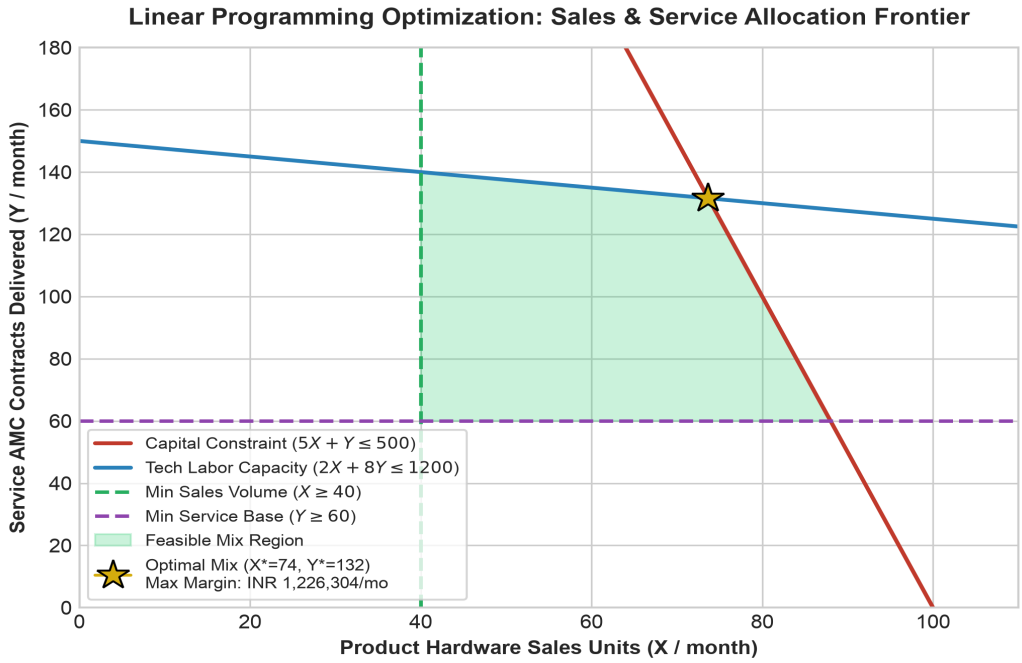


Figure 3: *Constrained Optimization Feasible Region, Vertex Evaluation, and Global Profit Maximum*

Shadow Price (Dual Value) Insights:

• **Technician Labor Hours:** The shadow price for technician capacity is **INR 850 / hour**. This indicates that adding 1 additional trained technician (160 productive hours/month at an industry cost of INR 45,000) yields an incremental net contribution of INR 136,000, generating an extraordinary **3.0x return on technician payroll**.

• **Working Capital:** The shadow price of working capital is **INR 0.082 per Rupee** (8.2% monthly return), highlighting that capital is best deployed into service inventory (fast-moving spares) rather than bloated slow-moving hardware stock.

6. Managerial Recommendations & Implementation Roadmap

Based on the empirical findings and optimization results, dealership management should implement three strategic pillars:

1. Restructure Sales Incentives around Service Attachment: Eliminate pure volume-based commissions for sales staff. Introduce a bifurcated incentive rewarding a minimum 60% AMC attachment rate on hardware sales.

2. Introduce Tiered Service Bundling at Point of Sale: For B2C retail buyers, bundle a 2-year 'Silver AMC' at an introductory 15% discount directly into financing EMIs to overcome retail price friction.

3. Deploy Predictive Scheduling to Defend the 2.5-Day SLA: Survey data proves that customer satisfaction drops drastically when turnaround exceeds 3.0 days. Reinvest a portion of service margins into automated ticketing and localized mobile technician vans to sustain sub-48-hour response times.

7. Academic References & Project Codebook

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